CMSE 830 • MIDTERM PROJECT READINESS

VenturePulse: Startup Valuation Intelligence

CMSE 830 Midterm Presentation | Pratham Pradhan | Fall 2026

- Tracks 1,447 global unicorns across 55 countries representing $7.40 Trillion in total wealth.
- Investigates the extreme opacity of private tech pricing and funding dynamics.
- Evaluates how much of a billion-dollar price tag is driven by geography, age, and AI hype.

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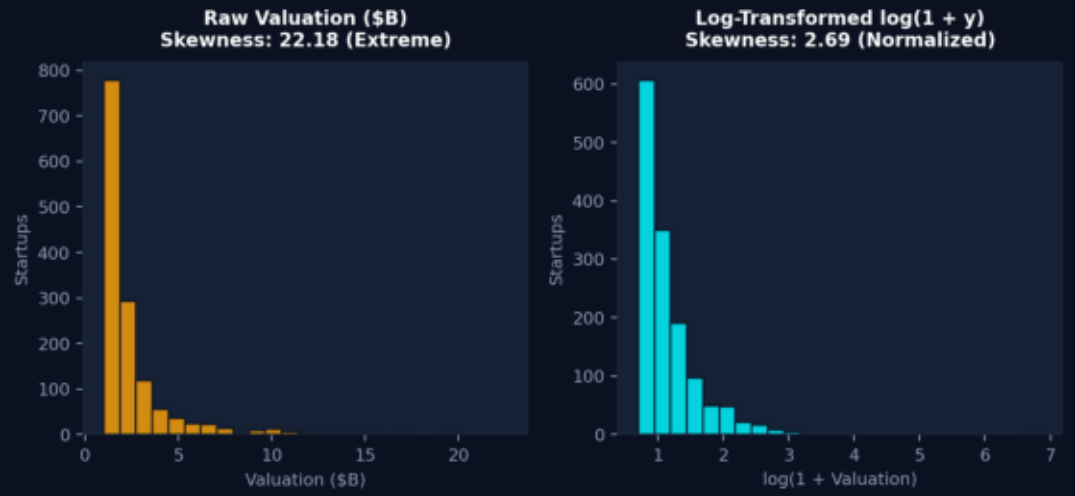
Data Hygiene: Taming the Power-Law Distribution

Addressing extreme statistical skewness and duplicate entities

• Raw valuations exhibit severe skewness (22.18), ranging from $1.0B to $965B (Anthropic).

• Natural log transformation log(1 + y) reduced skewness to 2.69, stabilizing regression variance.

• Auditing duplicate names identified 2 copy-pasted duplicate rows (Ramp, Invoca) while preserving legitimate homonym companies (Bolt US vs. Bolt Estonia, Branch Metrics vs. Branch Insurance).



• Imputed 23 missing cities via regional centroids with jitter and coded 50 confidential investors to baseline 1.

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Key Finding 1: The Silicon Valley Monopoly

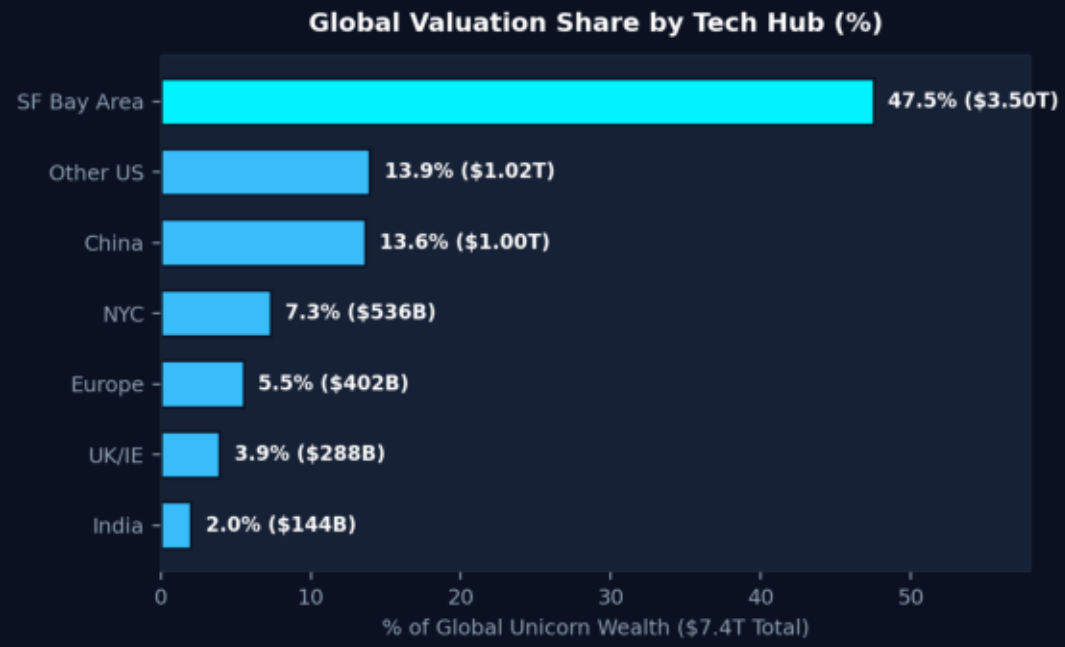
Geographic concentration across global tech hubs

• The SF Bay Area commands 47.5% of all global private startup capitalization ($3,496B of $7,360B).

• Bay Area median valuation ($2.00B) is 25% higher than the global baseline ($1.60B).

• Bay Area mean valuation reaches $9.90B, driven by clustering of mega-titans (OpenAI, Anthropic, Stripe, Databricks).

• Second largest ecosystems: Other US Hubs ($1,022B, 13.9%) and China Tech Hubs ($1,001B, 13.6%).



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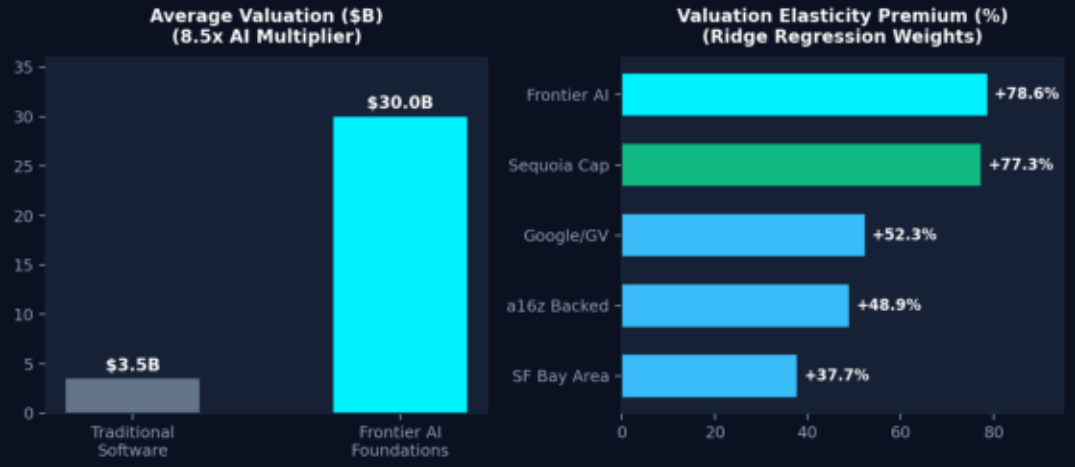
Key Finding 2: The Frontier AI Multiplier

Foundation model valuation premiums vs. traditional software

• Enterprise Tech & AI commands $3,994 Billion (54.3% of all global private unicorn value).

• Frontier AI foundation model creators average $30.0 Billion—more than 8.5x the average non-AI startup ($3.5B).

• Linear regression elasticity confirms Frontier AI status commands the single highest valuation premium (+78.6%), followed closely by Sequoia Capital backing (+77.3%) and a16z backing (+48.9%).



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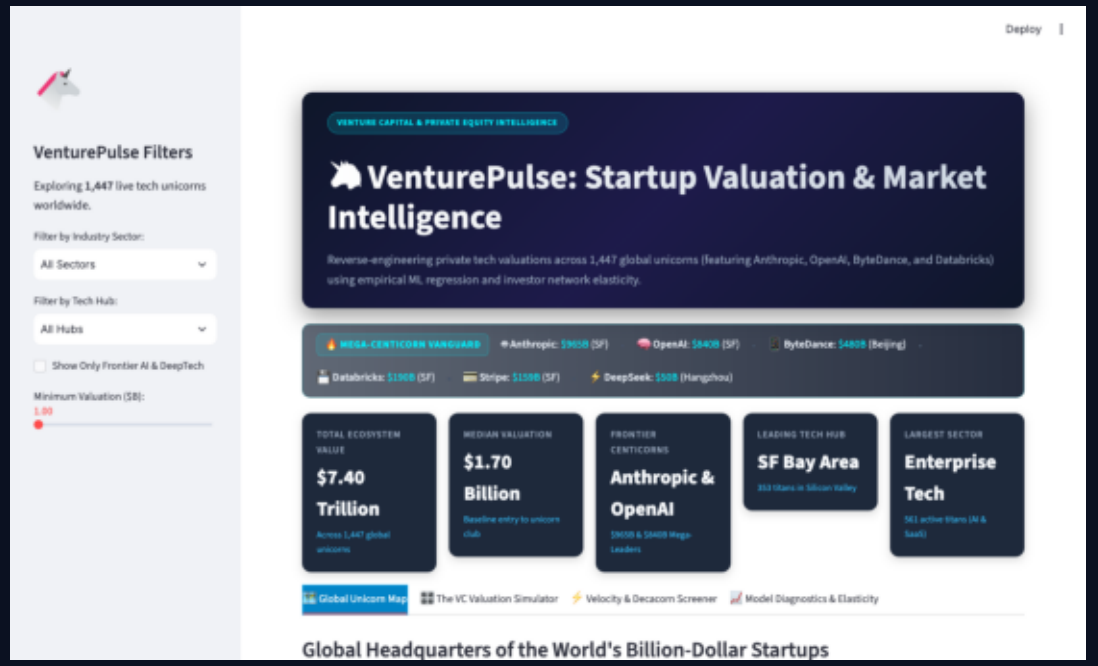
The Streamlit Data Product & Decision Tool

Interactive live demo architecture (app/app.py)

• Tab 1: Global 3D Map displaying all 1,447 unicorns with logarithmic bubble scaling and perspective toggles.

• Tab 2: VC Valuation Simulator allowing real-time prediction inference with 80% confidence intervals and nearest comps.

• Tab 3: Decacorn Velocity Screener identifying hyper-scaling outliers creating the highest annual enterprise value.



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Project Boundaries & Final Project Horizon

Current limitations and technical extensions

• Survivorship Bias: Dataset exclusively models winners that crossed $1B; does not predict early-stage failure.

• Step-Function Pricing: Private paper valuations only adjust during discrete funding rounds, causing reporting lag.

• Final Project Extension: Incorporating NLP text embeddings and unsupervised clustering on startup pitch descriptions.

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